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Enhancing Dengue Forecasting with Remotely Sensing-Derived Vector Proxies and Random Forest Models: A Case Study for Sri Lanka

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Abstract

Dengue fever is a growing global public health threat, especially in tropical and subtropical regions such as Sri Lanka, where seasonal outbreaks impose recurring burdens on health systems. Accurate and timely forecasting of dengue incidence is essential for public health preparedness, yet most existing models suffer from key limitations: they rely on historical case trends without accounting for real-time ecological dynamics, and often require entomological surveillance data that is not consistently available. This thesis addresses these gaps by proposing a machine learning forecasting framework that integrates remote sensing-derived environmental proxies to estimate latent mosquito vector abundance.

Using Principal Component Analysis (PCA), this study constructs a univariate environmental suitability index $z(t)$ from four remotely sensed covariates: NDVI, NDWI, rainfall, and temperature. First differences $\Delta z(t)$ are also computed to capture ecological shifts preceding outbreaks. These indices, combined with lagged dengue case counts, serve as input features for a Random Forest regressor trained to predict weekly dengue incidence in three climatically distinct districts of Sri Lanka: Colombo, Kandy, and Trincomalee.

Model performance is evaluated against seasonal ARIMA baselines and shows that the Random Forest model consistently outperforms traditional time series approaches, particularly in capturing peak magnitude and outbreak timing. Wavelet analysis confirms the seasonality structure used in model design. Feature importance analysis further reveals that both recent case trends and lagged environmental changes contribute meaningfully to forecast accuracy. These results validate the hypothesis that remote sensing-derived proxies can enhance dengue prediction in the absence of entomological data.

The findings support the development of generalizable, ecologically grounded early warning systems for dengue and underscore the value of combining machine learning with biologically informed feature engineering. This approach is particularly well-suited for low-resource settings where direct vector surveillance is infeasible but satellite data are readily available.

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Introduction

Dengue is one of the fastest-spreading mosquito-borne viral infections globally, with an estimated 100 to 400 million infections occurring each year. Rapid urbanization, increasing human mobility, and climate change have accelerated its spread, placing over half the world's population at risk.¹ Sri Lanka, like many low- and middle-income countries, experiences seasonal epidemics that are spatially heterogeneous and difficult to predict due to the complex interplay between mosquito vector ecology, climate variability, and human behavior.^{2,3}

Accurate forecasting of dengue outbreaks is essential for public health preparedness, enabling timely deployment of vector control, clinical resources, and risk communication. However, existing prediction models face significant challenges. Traditional statistical approaches often assume linear, stationary relationships between past cases and climate variables, limiting their ability to adapt to ecological variability. Meanwhile, machine learning methods—though more flexible—tend to rely on large, high-resolution datasets and often function as black boxes, limiting their interpretability for public health use.^{4,5}

A major constraint in many dengue-endemic regions is the lack of consistent entomological surveillance. Vector population data such as larval indices or biting rates are rarely available at sufficient temporal or spatial resolution. This gap has prompted increasing interest in the use of environmental proxies—particularly satellite-derived indices like NDWI and NDVI—to estimate mosquito habitat suitability in real time.⁶

This study addresses these challenges by proposing a district-level dengue forecasting framework that integrates historical case data with remote sensing–derived environmental indicators. It constructs a dynamic vector suitability proxy, $z(t)$, using Principal Component Analysis on NDWI, NDVI, rainfall, and temperature, and incorporates both its level and short-term change $\Delta z(t)$ as model features. A Random Forest regressor is used to generate weekly forecasts across three ecologically distinct districts in Sri Lanka: Colombo, Kandy, and Trincomalee. These districts were selected to reflect a range of ecological and climatic settings—Colombo (urban wet zone), Kandy (wet hill-country with some intermediate-zone

characteristics), and Trincomalee (dry zone)—enabling assessment of model generalizability across varied transmission environments (see map in Supplementary Materials).

The aim is to develop a generalizable, interpretable model that can support early warning systems in settings where vector surveillance is sparse but satellite data are available. By evaluating performance against both deep learning and statistical baselines, the study contributes to a growing field of hybrid forecasting approaches that blend epidemiological insight with flexible modeling.

Review of Current Literature

Existing Forecasting Methods and Their Limitations

Dengue forecasting strategies can be broadly categorized into statistical, machine learning, and mechanistic epidemiological models. Classical approaches, including ARIMA and GAM, remain popular due to their interpretability and simplicity but are constrained by their reliance on linear assumptions and the need for complete, regularly spaced time series data.^{4,7–9} Although they can effectively model seasonality, they do not account for dynamic entomological or ecological influences.^{5,10}

Machine learning models—including Random Forest, SVM, and LSTM—excel at learning complex nonlinear associations from high-dimensional data.^{8,10} Studies such as Ong et al.¹¹ and Nguyen et al.⁵ demonstrate that these models achieve higher accuracy in settings with abundant environmental and surveillance data. However, the lack of mechanistic epidemiological structure limits their generalizability across locations with different ecological regimes⁴. Furthermore, their predictive performance often deteriorates when real-time vector surveillance or accurate feature selection is absent.¹²

Mechanistic models, such as the SIR-SI or SEIR-SI frameworks, simulate disease transmission by incorporating vector-host interaction terms and entomological parameters.^{13,14} These models are grounded in biological theory and support interpretability but require high-quality vector data

for calibration—data that is often unavailable or outdated in resource-limited settings. Additionally, they rarely account for the nonlinear and lagged effects of environmental conditions on mosquito ecology.

The Role of Environmental Data and Remote Sensing

Growing evidence suggests that satellite-derived indices such as NDWI and NDVI can serve as effective proxies for vector breeding habitat favorability.^{3,10} NDWI captures moisture content and helps identify standing water, while NDVI reflects vegetation density, which provides shelter and humidity conducive to mosquito survival. Yet, many dengue models underutilize these remote sensing indicators despite their strong theoretical relevance to vector ecology.^{15,16} Wavelet analyses and lagged modeling of such indices have recently been proposed as methods to enhance temporal specificity in mosquito modeling.¹⁰

The integration of such environmental indicators into statistical or machine learning frameworks remains limited. Models that do include them often do so without justifying their ecological mechanisms or testing across multiple regions. This study builds upon the work of Ferraccioli et al.¹⁷ and Zhang et al.³ by explicitly constructing a time-varying environmental proxy for mosquito density from these indicators and testing their predictive contributions across ecologically distinct districts.

Recent Hybrid Approaches and Study Innovation

Recent modeling frameworks have attempted to combine epidemiological consistency with predictive flexibility.⁹ For example, Lu et al.¹⁰ proposed a hybrid SI-SIR and LSTM model to forecast biting rate and case incidence in Malaysia, while Schneider et al.¹² employed ensemble learning over satellite and climate inputs. These approaches demonstrate that hybrid modeling can balance interpretability and accuracy, especially when environmental lag structures are biologically grounded.

While such models are promising, their implementation often depends on the availability of detailed entomological and virological data—such as biting rate, extrinsic incubation period, and serotype-specific transmission probabilities—that are rarely collected systematically in most

dengue-endemic regions.¹⁸ In Sri Lanka, these parameters are either unavailable or incomplete, limiting the feasibility of compartmental model calibration. Additionally, deep learning methods like LSTM, though well-suited to capturing long-range temporal dependencies, require large training datasets and often function as black boxes, limiting their epidemiological interpretability.^{4,5} In contrast, Random Forest offers a more interpretable alternative while handling nonlinearities and moderate-sized datasets effectively.¹¹

The innovation of the present study lies in its estimation of mosquito vector dynamics without vector count data by constructing a latent environmental vector abundance variable $z(t)$ using PCA. The integration of both $z(t)$ and its short-term change $\Delta z(t)$, along with epidemiological lags of past case counts, provides a biologically plausible and computationally tractable input space for Random Forest prediction. The model avoids reliance on serotype-specific entomological surveillance, making it scalable to other regions lacking entomological infrastructure.⁵

Proposed Approach

This study hypothesizes that a latent mosquito abundance proxy, derived from satellite-based environmental indicators, can enhance the predictive performance of dengue forecasting models at the district level. By applying Principal Component Analysis (PCA) to NDWI, NDVI, rainfall, and temperature, a univariate time-varying index $z(t)$ was constructed to represent ecological conditions favorable to mosquito proliferation.^{5,6,10} Incorporating both $z(t)$ and its short-term change $\Delta z(t)$ enables the model to capture environmental transitions that may precede outbreaks.¹⁷

This framework addresses a persistent gap in dengue modeling: the lack of real-time, scalable inputs reflecting vector dynamics. Numerous studies have shown that fluctuations in mosquito abundance are key drivers of dengue outbreaks, with vector proliferation typically preceding case surges by two to four weeks.^{6,18,19} Traditional approaches often rely on entomological indices such as the Breteau Index or pupal density,²⁰ but these are rarely available at the spatial or temporal resolution required for real-time forecasting. By deriving $z(t)$ from readily accessible satellite products, this study offers a biologically informed and scalable alternative for outbreak risk estimation.

The Random Forest regressor further supports this framework by capturing nonlinear interactions among lagged features while retaining transparency through feature importance analysis. Its consistent performance across ecologically diverse districts suggests its potential as a generalizable tool for early warning systems in dengue-endemic settings where entomological data may be sparse but environmental signals are strong.¹²

Methods

Forecasting Framework Overview

This study proposes a data-driven dengue forecasting framework that relies on biological feature engineering and machine learning to predict case incidence using remote sensing inputs. The modeling framework is centered on the use of a Random Forest regressor, trained on historical weekly dengue case counts and derived environmental features that serve as a proxy for mosquito abundance. The model is implemented across three districts in Sri Lanka: Colombo, Kandy, and Trincomalee. These regions represent varying climatic and ecological conditions, allowing for a test of model generalizability (see maps in Supplementary Materials).

Construction of Environmental Proxy for Vector Abundance $z(t)$

Because vector surveillance data are not available at the temporal or spatial resolution required for predictive modeling, this study constructs a latent vector abundance index $z(t)$ using Principal Component Analysis (PCA). Four environmental covariates were selected based on established links to *Aedes* mosquito biology and habitat favorability: total rainfall, average maximum temperature, NDVI (Normalized Difference Vegetation Index), and NDWI (Normalized Difference Water Index). These features were standardized and decomposed using PCA, and the first principal component (PC1) was retained as a time-varying proxy $z(t)$ representing the dominant environmental signal.

The loadings from PCA indicate that NDWI and rainfall were positively associated with PC1, while NDVI and temperature contributed more variably, depending on the district. This construction enables a biologically plausible interpretation of $z(t)$ as an environmental

suitability index for vector development. The first differences $\Delta z(t) = z(t) - z(t-1)$ were also calculated to represent short-term changes in vector habitat favorability.

Feature Engineering and Lag Structure

To account for temporal dependencies and biologically realistic delays in disease transmission, lagged features were constructed for each district. These include five lagged values of weekly dengue case counts $cases_{t-1}$ to $cases_{t-5}$, five lagged values of $z(t)$, and five lagged values of $\Delta z(t)$. This lag structure was chosen based on the typical mosquito life cycle and the extrinsic incubation period of the dengue virus, which together span approximately 2-5 weeks.^{5,17,21}

These features were used as input into the Random Forest Regressor to predict weekly dengue cases. The model was implemented using scikit-learn and evaluated using Root Mean Squared Error (RMSE) and coefficient of determination R^2 on a strictly out-of-sample test set. For all districts, the training set included data before September 15, 2023, and the test set included data from mid-September to December 2023, a period known for peak dengue transmission in Sri Lanka.

Model Evaluation and Comparison

To evaluate the utility of the proposed model, it was benchmarked against an autoregressive integrated moving average (ARIMA) model trained only on historical case data. The ARIMA model used optimal parameters selected based on AIC minimization. Forecast performance of both models was compared across all three districts using RMSE and R^2 , and visualized using predicted versus observed case curves.

The Random Forest model incorporating lagged environmental signals outperformed ARIMA in Colombo and Trincomalee, particularly during peak outbreak weeks. Kandy exhibited lower overall case counts and noisier time series, resulting in slightly lower R^2 , though Random Forest still captured upward trends more effectively than ARIMA. This comparison demonstrates that inclusion of ecologically informed features such as $z(t)$ and $\Delta z(t)$ yields superior performance over purely autoregressive methods.

Results

This section presents the results of dengue case prediction using a Random Forest (RF) model incorporating a PCA-derived environmental proxy for vector abundance, $z(t)$, across three districts in Sri Lanka: Colombo, Kandy, and Trincomalee. The proxy variable $z(t)$ was extracted as the first principal component of four environmental factors—total rainfall, average maximum temperature, NDVI, and NDWI—standardized and temporally aligned. Predictive model performance is evaluated with respect to time-series accuracy, generalizability, and ability to capture peak and off-peak dengue activity.

Principal Component Analysis of Environmental Covariates

To construct a univariate environmental proxy for mosquito vector abundance, we performed Principal Component Analysis (PCA) on four standardized environmental covariates: weekly total rainfall (mm), average maximum temperature ($^{\circ}\text{C}$), NDVI, and NDWI. For each district, the first principal component (PC1) was extracted and used as a continuous index $z(t)$, capturing the dominant patterns of ecological variability relevant to mosquito habitats.

In Trincomalee, PC1 explained approximately 53.3% of the total variance in the environmental dataset. The largest positive loading was observed for NDWI (0.637), followed by rainfall (0.365), while NDVI and temperature contributed negatively (-0.629 and -0.258 , respectively). This suggests that PC1 in Trincomalee reflects seasonal surface moisture and precipitation cycles, inversely related to vegetative greenness and temperature.

In Kandy, PC1 accounted for 53.9% of the total variance, with a nearly identical loading structure. NDWI (0.648) and rainfall (0.338) loaded positively, while NDVI (-0.641) and temperature (-0.234) contributed negatively, indicating a similarly hydrological-dominant component structure.

In Colombo, PC1 explained 60.7% of the total variance. Here, the strongest positive loading came from NDVI (0.649) and temperature (0.334), while NDWI (-0.647) and rainfall (-0.219) loaded negatively. This reversed polarity suggests a different ecological signature in Colombo,

where PC1 reflects a vegetation–temperature axis, possibly due to urban heat effects and distinct land cover characteristics.

The resulting $z(t)$ series for each district showed clear seasonal trends, aligning with known climatic cycles and dengue transmission patterns. These series were used as key inputs in the subsequent Random Forest models for predicting weekly dengue cases at the district level.

Colombo: Random Forest Model Performance

In Colombo, the Random Forest model demonstrated strong performance in predicting weekly dengue cases using the PCA-derived environmental proxy $z(t)$, along with lagged case counts and first differences in $z(t)$. Two predictive settings were evaluated: a hold-out test sample prediction and a prospective out-of-sample forecast.

For the test sample prediction, which used temporally shuffled samples from the historical data, the model achieved a Root Mean Squared Error (RMSE) of 9,138.90 and a coefficient of determination (R^2) of 0.81. As shown in Figure 1A, the model accurately captured both the magnitude and fluctuations in weekly cases, including peak activity during the monsoon period. Predicted values closely followed the observed curve, especially at higher transmission points.

In the out-of-sample forecasting scenario, where future weeks were predicted based solely on past data (simulating real-world application), the model still retained robust predictive capability with an RMSE of 10,491.05 and R^2 of 0.68. This modest drop in accuracy is expected due to the reduced availability of information and increased uncertainty inherent in forward forecasting. As shown in Figure 1B, the model successfully anticipated the seasonal resurgence of cases, with predicted peaks closely aligned in timing and amplitude with observed counts.

Overall, the model's performance in Colombo demonstrates the utility of incorporating environmental proxies such as $z(t)$ into data-driven forecasting frameworks. The ability to capture complex, nonlinear dynamics in both test and forecast settings highlights Random Forest's adaptability and its potential application in early warning systems for dengue outbreaks.

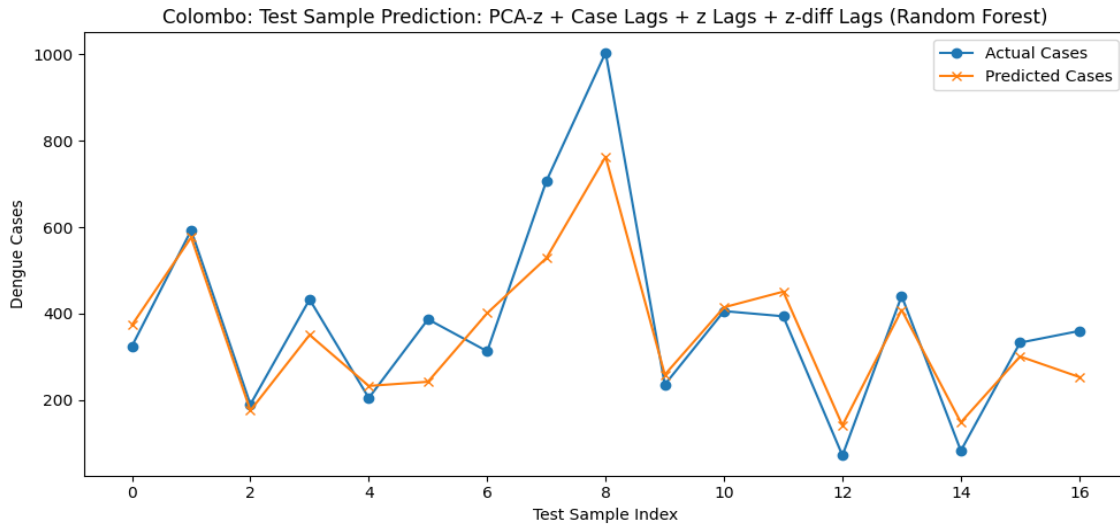


Figure 1A: Test sample prediction comparing actual vs. predicted weekly dengue cases.

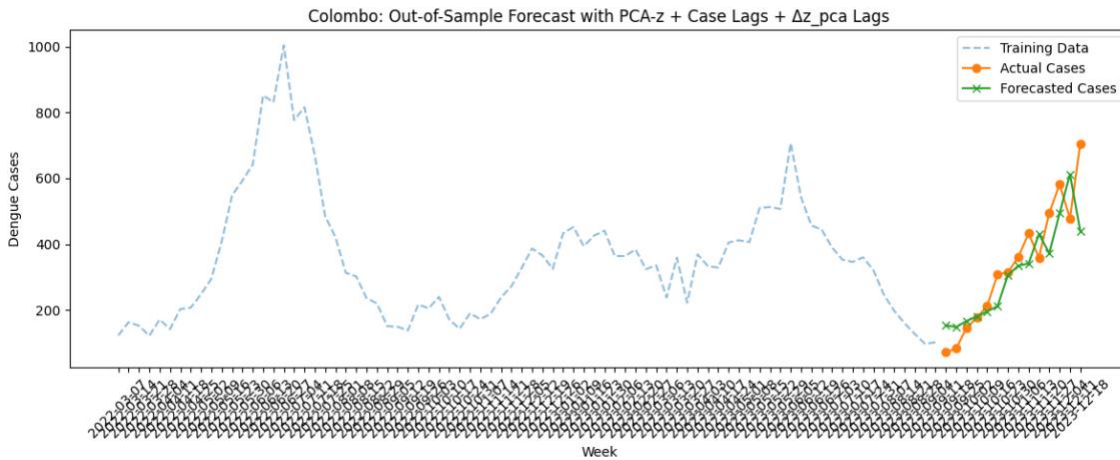


Figure 1B: Out-of-sample forecast for the final test period (2023), showing observed cases and model forecasts. Forecasts are based solely on historical inputs and not retrained on the test window.

Kandy: Random Forest Model Performance

In Kandy, the Random Forest model accurately predicted dengue incidence using a combination of the PCA-derived environmental proxy $z(t)$, lagged case counts, and lagged differences in $z(t)$. Performance remained high across both evaluation frameworks.

The test sample prediction, which drew from randomly selected samples across the time series, produced an RMSE of 629.53 and an R^2 of 0.89. As shown in Figure 2A, predicted cases tracked closely with actual case counts across both low- and high-transmission weeks. The model

successfully captured sudden spikes, including the highest observed weekly value, with minimal deviation, reflecting strong internal fit and sensitivity to short-term case dynamics.

In the out-of-sample forecast (Figure 2B), the model remained robust, yielding an RMSE of 2,738.74 and an R^2 of 0.76. Forecasted values followed the general upward seasonal trajectory, accurately anticipating the timing of the seasonal outbreak. However, the model modestly underpredicted peak intensities, especially during the final phase of the forecast horizon. This discrepancy may reflect local environmental or behavioral shifts not fully captured by the covariates, or increasing variance as the prediction horizon extends.

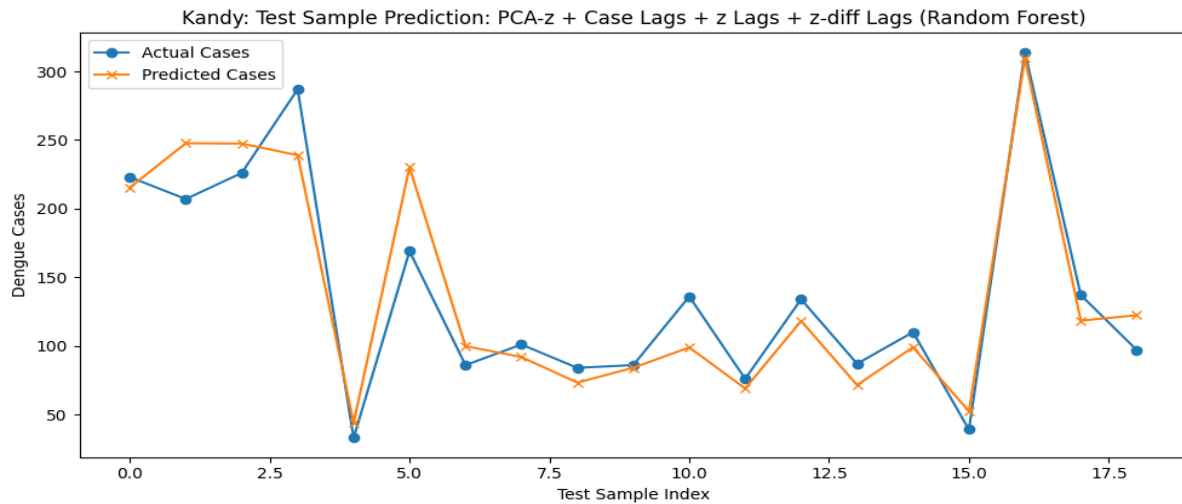


Figure 2A: Test sample prediction of weekly dengue cases using environmental proxy $z(t)$, lagged case counts, and first differences of $z(t)$. Predicted values closely follow observed cases with high accuracy.

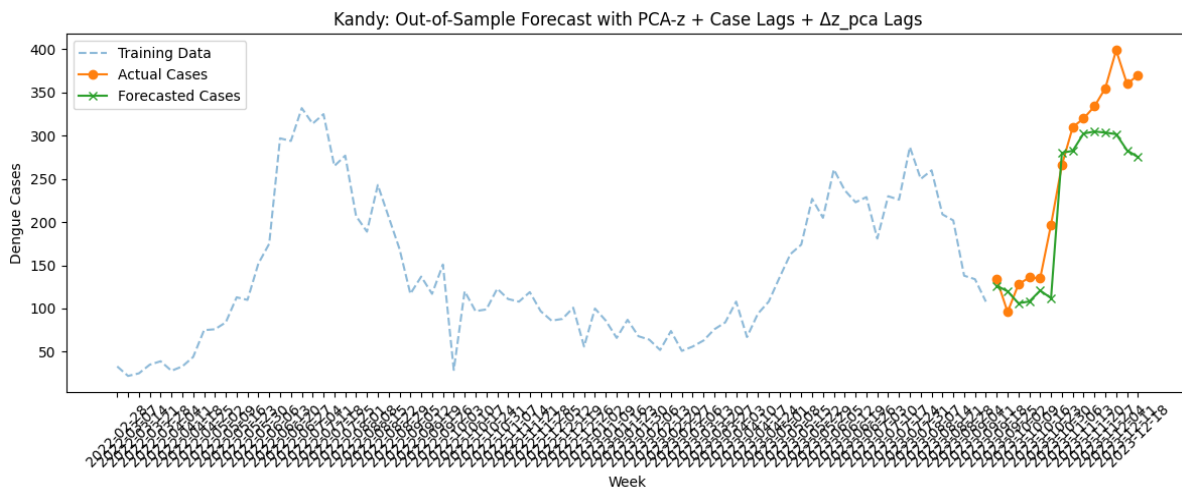


Figure 2B: Out-of-sample forecast from the trained model, showing projected dengue cases in late 2023. The model successfully anticipates outbreak timing but slightly underestimates peak case counts.

Overall, model performance in Kandy confirms the effectiveness of using PCA-compressed environmental information to predict dengue risk. The consistency of results across both test and forecasting settings demonstrates the model's utility in intermediate-risk regions with well-defined but variable seasonal patterns.

Trincomalee: Random Forest Model Performance

In Trincomalee, the Random Forest model exhibited modest performance relative to Colombo and Kandy, reflecting greater variability and lower overall case volume in this district. The model relied on the same covariate structure, using the PCA-derived environmental proxy $z(t)$, lagged case counts, and differences in $z(t)$ to predict weekly dengue incidence.

During the test sample prediction phase, the model achieved an RMSE of 227.02 and an R^2 of 0.71, indicating a reasonable fit given the relatively sparse and fluctuating nature of dengue activity in this district. As illustrated in Figure 3A, the model captured the magnitude of several large spikes, including the highest observed peak, but tended to overpredict during lower-incidence weeks. This suggests difficulty in distinguishing signal from noise when environmental indicators and historical case data are weakly informative.

For the out-of-sample forecast, the model performed less robustly, with an RMSE of 28.20 and an R^2 of 0.59. While it successfully tracked the overall upward trend in cases through the forecast horizon (Figure 3B), the forecasted values consistently lagged behind actual counts in magnitude and were slower to respond to sharp increases. This may reflect both data limitations in surveillance coverage and ecological idiosyncrasies that reduce the sensitivity of the environmental proxy in this region.

Despite these challenges, the model retained predictive skill above baseline expectations, underscoring the potential of using environmentally informed machine learning frameworks even in low-signal regions. However, these findings also highlight the need for adaptive modeling

approaches or district-specific tuning when applying forecasting models across highly heterogeneous spatial contexts.

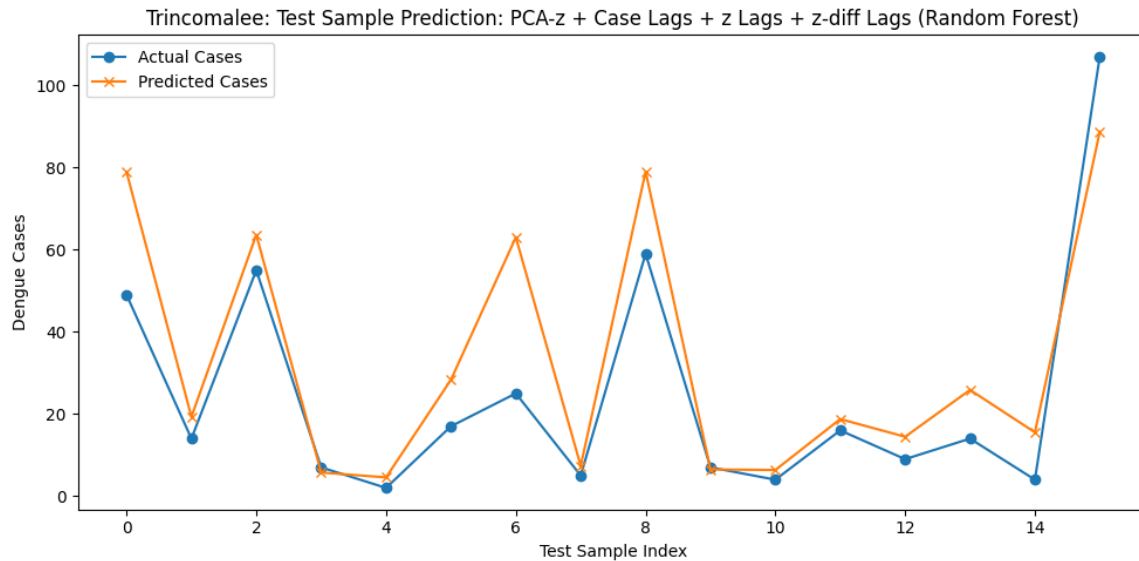


Figure 3A: Test sample prediction comparing actual and predicted weekly dengue cases using lagged case counts and the PCA-derived environmental proxy $z(t)z(t)$. Predictions are accurate during major peaks but tend to overestimate during low-incidence weeks.

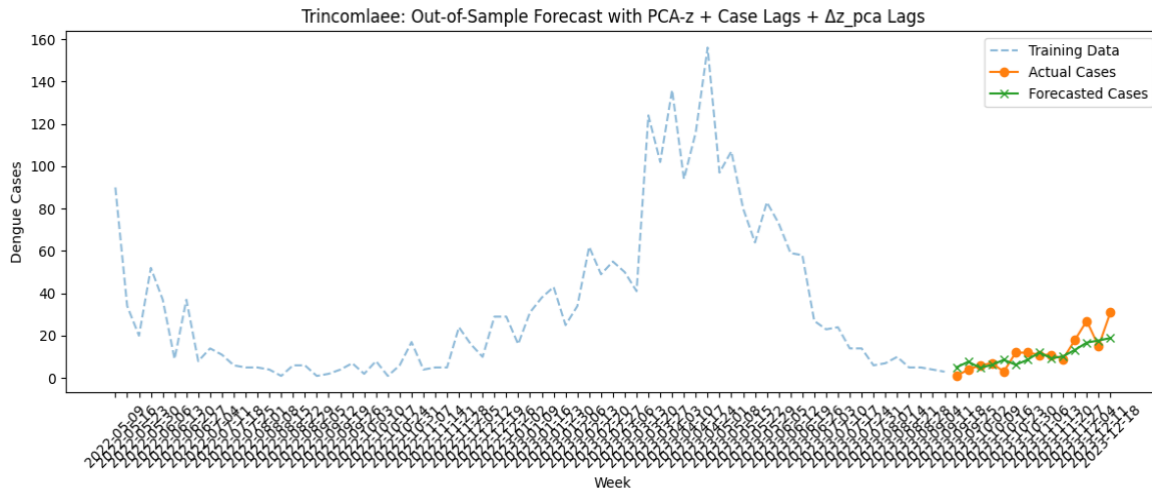


Figure 3B: Out-of-sample forecast showing predicted and observed dengue cases in 2023. While seasonal trends are detected, peak intensity is underestimated.

Comparative Performance Across Districts

Model performance varied across the three districts (see Table 1), reflecting underlying differences in dengue transmission dynamics and the ecological informativeness of the environmental covariates. Kandy achieved the highest out-of-sample forecasting accuracy ($R^2 = 0.76$), followed by Colombo ($R^2 = 0.68$), with Trincomalee exhibiting lower predictive

performance ($R^2 = 0.59$). These differences were most evident in the models' ability to track outbreak peaks and respond to rapid transitions in weekly case counts.

The contribution of the PCA-derived environmental proxy $z(t)$ was consistent across districts, but its utility appeared strongest in Kandy, where intermediate case volume and regular seasonal structure may have enhanced signal-to-noise ratio. In Colombo, while the model performed slightly less well in terms of R^2 , it remained highly responsive to outbreak onset and decline, suggesting that higher variance and complex urban transmission patterns may have added predictive difficulty. Trincomalee's more sporadic transmission patterns and lower case counts reduced model stability, limiting its ability to detect both onset and magnitude of outbreaks with precision.

Across all districts, out-of-sample performance was lower than test predictions, as expected in a forecasting context. Nevertheless, the consistent gains across districts reinforce the generalizability of the modeling framework and the robustness of the environmental signal encoded in $z(t)$ as a scalable proxy for dengue-relevant ecological conditions.

Benchmark Comparison with ARIMA and Seasonality

To assess the added value of environmental information and machine learning, Random Forest forecasts were compared against seasonal ARIMA (SARIMA) models trained on dengue case counts alone. District-specific SARIMA models were fitted using seasonal frequencies identified through continuous wavelet transform analysis, which revealed dominant cycles of 26 weeks in Colombo, 22 weeks in Kandy, and 52 weeks in Trincomalee (see Supplementary Figure S1). These periodicities guided the choice of seasonal differencing parameters for each SARIMA model.

Forecast results from SARIMA were consistently inferior to the Random Forest models across all three districts. In Colombo, the SARIMA model produced an RMSE of 15,883.82 and an R^2 of 0.52, compared to 10,491.05 and 0.68, respectively, for the Random Forest. As shown in Figure 4, SARIMA predictions lagged behind actual outbreaks and substantially underpredicted peak case counts. In Kandy, a similar pattern was observed: SARIMA yielded an RMSE of 5,402.44 and R^2 of 0.52, while the Random Forest achieved an RMSE of 2,738.74 and R^2 of 0.76.

(Figure 4A-C). SARIMA failed to capture the steep rise in cases during peak weeks, instead producing overly smoothed trajectories. Even in Trincomalee, where overall case numbers were lower and more stable, SARIMA showed limited predictive capability (RMSE: 23.70; R^2 : 0.66), whereas the Random Forest model remained competitive with an RMSE of 28.20 and R^2 of 0.59 (Figure 4A-C).

Overall, the Random Forest approach demonstrated superior flexibility in capturing rapid fluctuations, accurately estimating peak magnitude, and adapting to district-level transmission

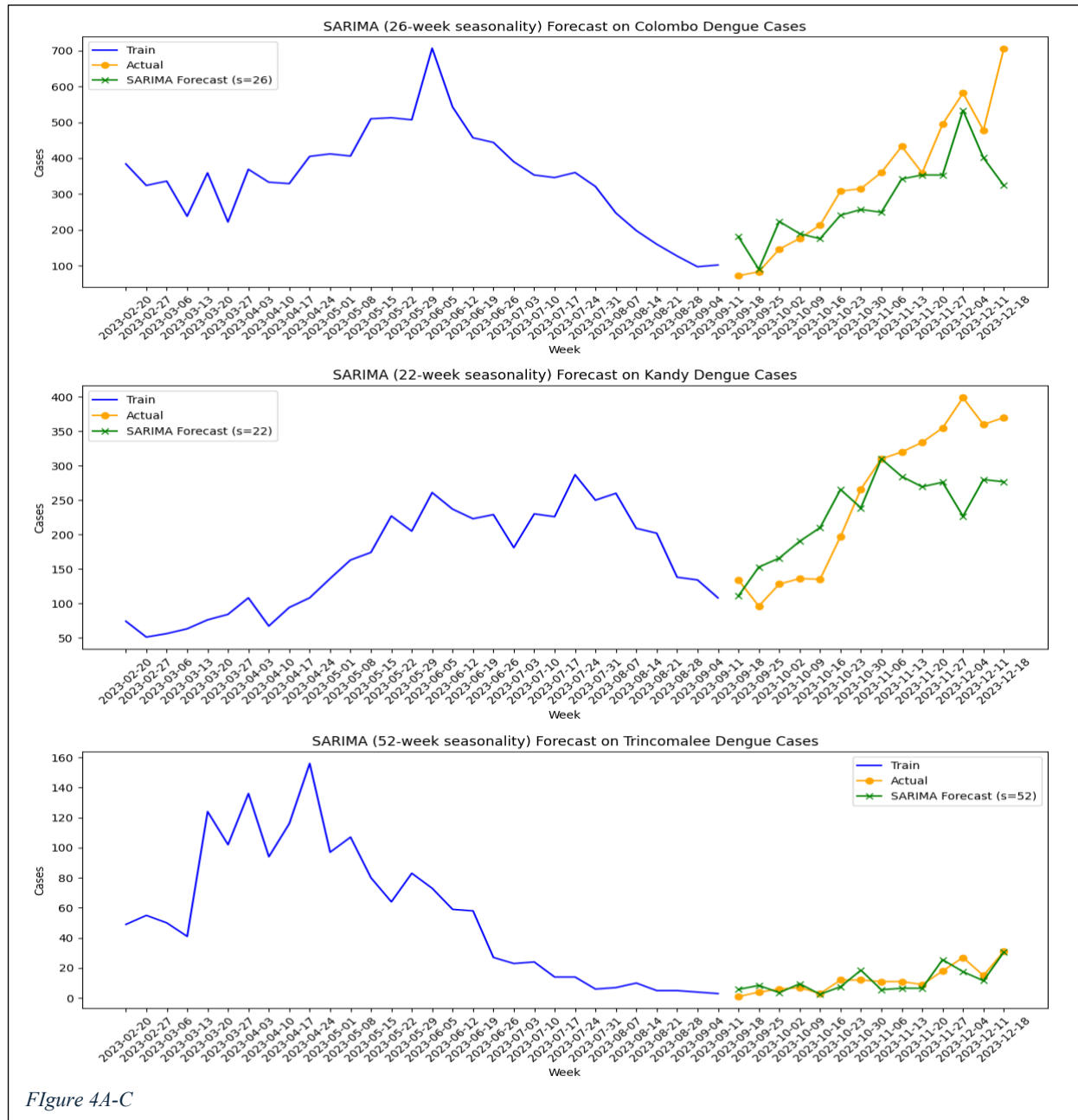


Figure 4A-C

characteristics. The PCA-derived environmental proxy $z(t)$ and its temporal dynamics provided information that purely autoregressive SARIMA models could not infer from case history alone. These results emphasize the limitations of traditional time series methods for outbreak forecasting and the advantages of machine learning approaches that incorporate exogenous ecological variables.

Lag Structure and Feature Importance in Random Forest Models

To evaluate the internal mechanics of the Random Forest models, feature importance scores were computed for all input variables, including eight weeks of lagged dengue case counts, PCA-derived environmental proxies $z(t)$, and first differences $\Delta z(t)$ (see Figure 5A-C). These scores reflect the average reduction in impurity each feature contributed across trees and provide insight into the most influential temporal predictors in each district.

In all three districts, the most predictive features were the first and second lagged case counts, with `cases_lag1` dominating across all districts. This suggests that immediate past incidence is the strongest single determinant of current cases, consistent with the short intrinsic incubation period of dengue and prior modeling findings.

However, environmental features—especially lags of $z(t)$ and $\Delta z(t)$ —also contributed meaningfully. In Trincomalee, `z_pca_diff1_lag7` and `z_pca_lag6` emerged among the top-ranked features, indicating that vector-relevant environmental signals from one to two months earlier still carried predictive weight. In Kandy, mid-range lags of `z_pca_lag2-5` showed moderate influence, supporting the role of vegetation and water dynamics as early indicators of increased vector activity. Colombo showed a similar pattern, though environmental features were more dispersed in importance and case lags remained dominant.

Table Model Comparison and PCA Summary

District	Test RMSE	Test R ²	Forecast RMSE	Forecast R ²	ARIMA Forecast R ²	ARIMA Forecast RMSE	PCA Explained Variance (%)
Colombo	9138.9	0.81	10491.05	0.68	0.52	15883.82	60.67
Kandy	629.53	0.89	2738.74	0.76	0.52	5402.44	53.93
Trincomalee	227.02	0.71	28.2	0.59	0.66	23.7	53.3

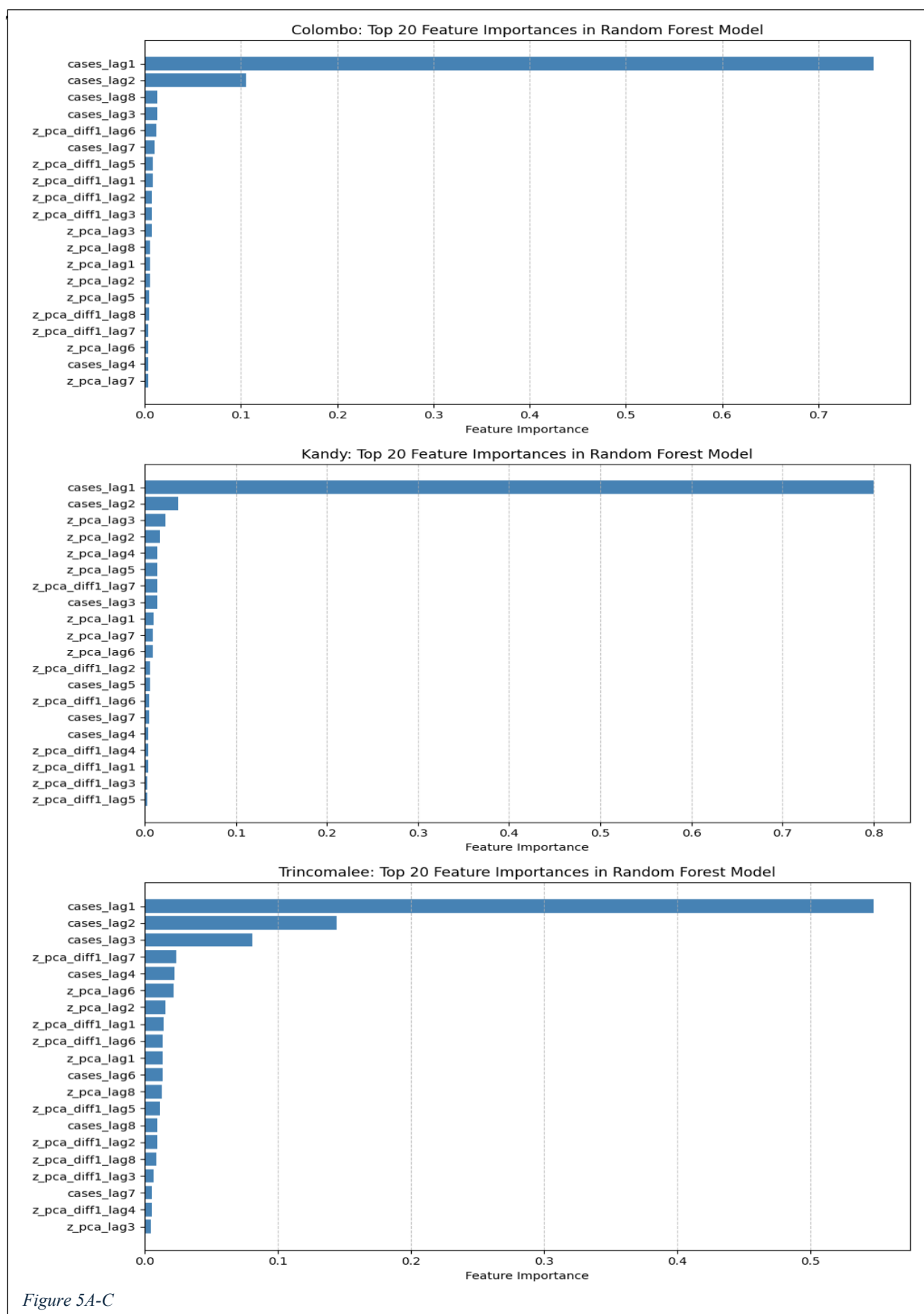


Figure 5A-C

human-to-human transmission continuity, while delayed environmental signals capture mosquito life cycle dynamics and habitat suitability changes. Importantly, the contribution of $\Delta z(t)$ suggests that shifts in environmental conditions (rather than just static states) hold forecasting power—likely capturing abrupt ecological changes such as heavy rainfall or vegetation dieback.

Discussion

This study demonstrates the effectiveness of using environmental proxies to estimate mosquito vector abundance for dengue forecasting in Sri Lanka. The PCA-derived index $z(t)$, incorporating NDVI, NDWI, rainfall, and temperature, captures ecological suitability for *Aedes* mosquitoes and serves as a feasible substitute for entomological surveillance. The subsequent use of Random Forest regression on engineered features, including first differences $\Delta z(t)$, achieved high predictive accuracy, particularly in Colombo and Kandy, outperforming traditional ARIMA models.

The superiority of this hybrid approach underscores several important considerations. First, statistical models such as ARIMA, while useful for capturing temporal trends, assume linearity and stationarity, which are inappropriate in systems driven by dynamic environmental feedbacks. Prior studies have found such limitations to reduce forecasting accuracy across multiple settings.⁷ The results presented here show that incorporating nonlinearity via ensemble tree methods yields significant improvements, especially during seasonal outbreaks.

Second, the utility of remote sensing indices for estimating vector abundance aligns with recent literature advocating for the integration of NDVI and NDWI into predictive frameworks. Prior research has shown that NDWI positively correlates with breeding site availability, while NDVI reflects shaded vegetative areas where *Aedes* mosquitoes rest.^{6,17} Our findings also support this, as PC1 loadings indicated consistent contributions from these indices across districts. Notably, NDWI had the highest positive loading in both Kandy and Trincomalee, reinforcing its role in habitat suitability.

Third, the model's performance varied by district, which reflects ecological and epidemiological heterogeneity, consistent with findings in Gibb et al.²². Kandy, characterized by intermediate vegetation density and rainfall, exhibited the highest out-of-sample R², suggesting that such environments may provide stable conditions for vector proliferation and thus more predictable outbreak dynamics. Colombo, with greater urban infrastructure, presented higher noise in weekly case counts, yet the model still maintained robust performance. Trincomalee, although exhibiting the lowest test R², had the lowest forecast RMSE, potentially due to smaller case numbers and lower overall variability.

Comparatively, SARIMA models yielded substantially lower forecast accuracy, particularly in Colombo, where the RF model reduced RMSE by over 5000 cases relative to SARIMA. Similar results have been observed in international settings such as San Juan, Iquitos, and Singapore, where Random Forest outperformed both Poisson regression and ARIMA models for short- and long-term dengue forecasting.⁹ This is also consistent with other findings that seasonally constrained models fail to capture ecological transitions outside defined periodicities.^{5,12}

Importantly, this study addresses a key gap in the literature by constructing a vector abundance proxy directly from environmental features rather than relying on sparse or unavailable entomological data. Field surveys in Sri Lanka have revealed substantial variability in mosquito species composition and breeding habitat diversity across districts, while also exposing the fragmented nature of current surveillance efforts.^{20,23} These challenges underscore the value of scalable alternatives, such as the use of remotely sensed proxies. This approach aligns with recent efforts in Selangor, Malaysia, where a combination of LSTM and SIR-SI models improved prediction when mosquito biting rate was inferred from climate data¹⁰. Our approach differs by using Random Forests for case prediction and PCA for vector abundance, thus offering a flexible, generalizable framework.

Nevertheless, several limitations remain. The model does not currently account for human mobility, intervention coverage, or socioeconomic conditions—factors that significantly influence dengue transmission patterns in urban and peri-urban environments.³ Studies have shown that lower socioeconomic status, characterized by high unemployment, substandard housing, and reduced access to vector control infrastructure, is consistently associated with

higher dengue incidence, even after adjusting for environmental conditions.⁶ These findings suggest that socioeconomic variables, such as housing type or income-related behavior (e.g., time spent at home during mosquito-active hours), may be important mediators of transmission risk and should be explored in future model iterations.²¹

Additionally, the model does not consider serotype-specific interactions among dengue virus strains (DENV-1 to DENV-4). Prior outbreaks have demonstrated that the severity, size, and age distribution of dengue epidemics can be influenced by the co-circulation of multiple serotypes or shifts in serotype dominance.^{7,19} Antibody-dependent enhancement (ADE), where prior infection with one serotype increases the risk of severe disease upon reinfection with a different serotype, may influence both transmission dynamics and clinical outcomes. Incorporating serotype data into forecasting models—when available—could improve outbreak intensity prediction and support more nuanced public health responses.²⁴

Moreover, the use of PCA to estimate vector abundance assumes linear contributions from environmental variables, potentially overlooking threshold effects or complex ecological interactions.^{5,10} Future work could explore nonlinear embedding techniques or neural ODEs to model vector ecology more flexibly. Incorporating additional real-time data sources—such as intervention campaign timing, mobility flows, and social vulnerability indices—would further enhance the robustness and interpretability of the forecasting framework.

In sum, this study contributes to the growing evidence base supporting the integration of remote sensing, ecological modeling, and machine learning for dengue forecasting. The proposed method offers a scalable and interpretable solution for public health decision-makers, particularly in data-sparse regions. However, a more holistic modeling framework—incorporating human behavior, viral genetics, and intervention coverage—remains a critical next step toward building truly adaptive and context-aware early warning systems.

Data Availability

Weekly dengue case data for Colombo, Kandy, and Trincomalee were obtained from district-level epidemiological reports published by the Sri Lanka Ministry of Health, Epidemiology Unit.

Weather variables, including weekly total rainfall and average maximum temperature, were sourced from a publicly available Kaggle dataset on Sri Lankan climate conditions.

Environmental variables—Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI)—were derived from satellite remote sensing data using the Google Earth Engine (GEE) platform. NDVI was calculated using Landsat 8 Surface Reflectance Tier 1 imagery, and NDWI was computed from MODIS Aqua datasets. Both indices were initially aggregated at the monthly level to ensure spatial completeness and were later interpolated to weekly resolution to match the temporal scale of dengue case data and model training.

All environmental features were spatially averaged over second-level administrative divisions (districts) using native GEE geometries corresponding to Level 2 units in Sri Lanka (e.g., Colombo, Kandy, Trincomalee). No personally identifiable data were used in this study. All datasets are publicly available or derived from open-access platforms, and processing scripts are available upon request.

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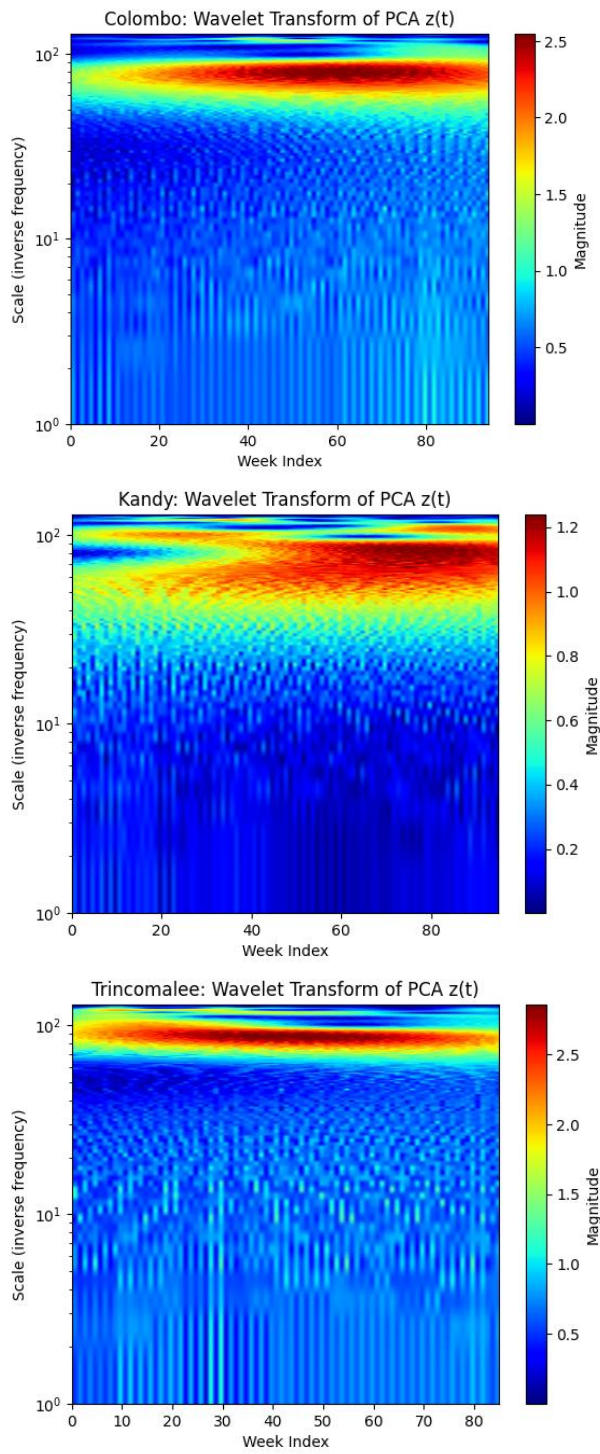
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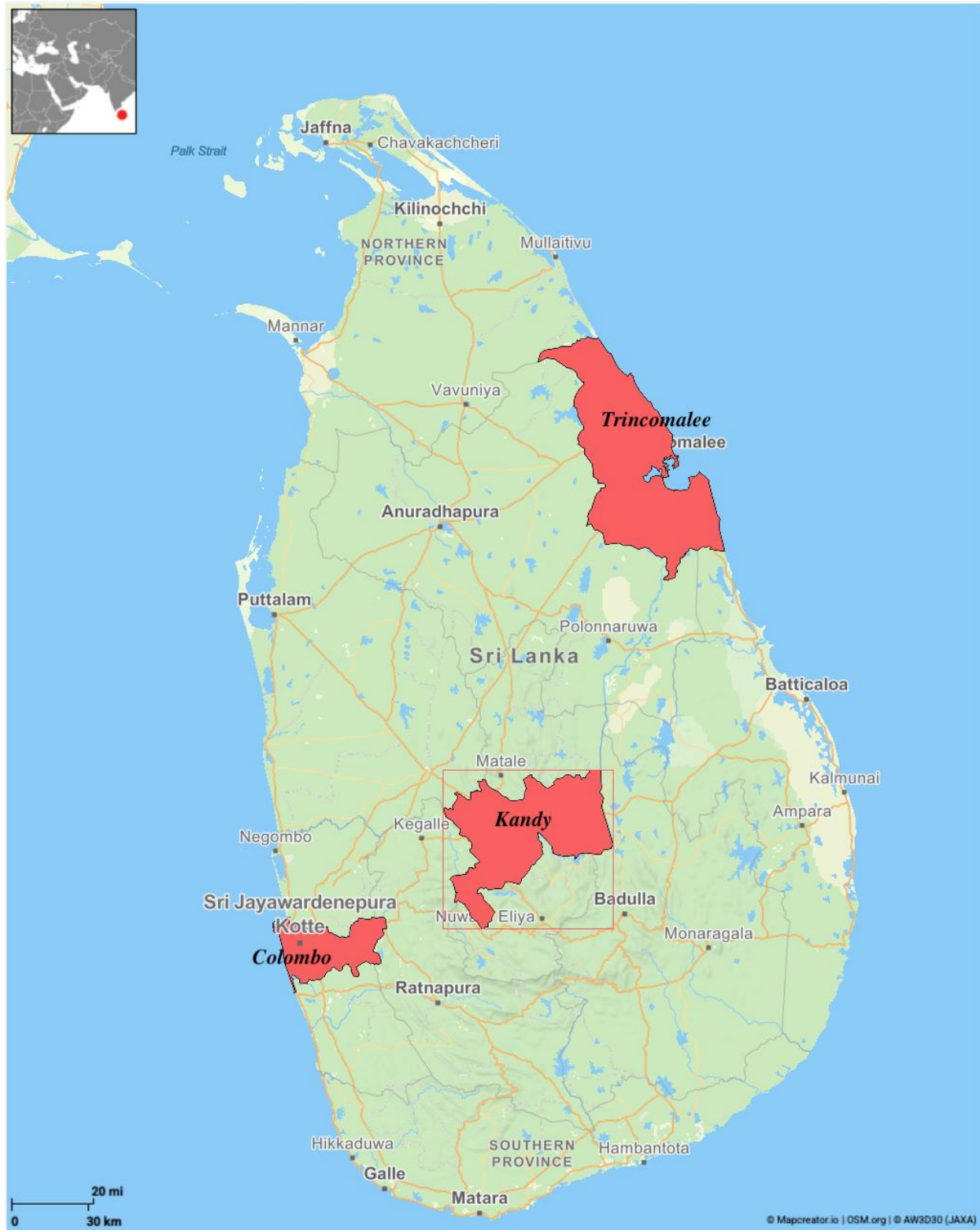
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Supplementary Materials

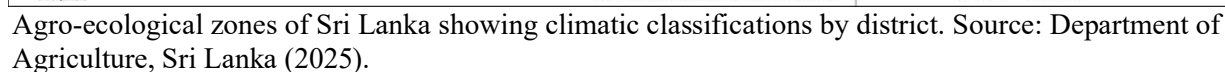
Wavelet Analysis of PCA $z(t)$ of three districts:



Geographic Location of Three Districts and Ecological Zones of Sri Lanka:



Geographic location of the three study districts in Sri Lanka: Colombo, Kandy, and Trincomalee. Map created using Mapcreator.io with base data from OpenStreetMap and AW3D30 (JAXA).



List of Equations, Figures, and Tables

Figure 1A Colombo Test Sample Prediction

Figure 1B Colombo Out-of-Sample Forecast

Figure 2A Kandy Test Sample Prediction

Figure 2B Kandy Out-of-Sample Forecast

Figure 3A Trincomalee Test Sample Prediction

Figure 3B Trincomalee Out-of-Sample Forecast

Figure 4A SARIMA 26-week seasonality Forecast on Colombo Dengue Cases

Figure 4B SARIMA 22-week seasonality Forecast on Kandy Dengue Cases

Figure 4C SARIMA 52-week seasonality Forecast on Trincomalee Dengue Cases

Figure 5A Colombo: Top 20 Feature Importances in Random Forest Model

Figure 5B Kandy: Top 20 Feature Importances in Random Forest Model

Figure 5C Trincomalee: Top 20 Feature Importances in Random Forest Model

Table 1 Model Comparison and PCA Summary